

Frontier-Tuned Procure-to-Pay Exception Investigator

Teach an enterprise agent to investigate ambiguous blocked invoices the way its best AP specialists do - using company evidence, workflows, tools, and controls.

1. Scenario

Routine ERP matching has blocked a high-value supplier invoice. The PO, invoice, goods receipt, contract amendment, quality hold, supplier email, and warehouse note disagree. The agent must determine the true root cause, gather the smallest sufficient body of evidence, and recommend a safe disposition.

Example: 4,800 cases physically arrived, but 1,200 failed ERP receipt because of a UOM mapping error and entered quality hold. Separately, a supplier surcharge is invalid because its acknowledgment arrived after dispatch. Retain the payment hold, propose a controlled receipt correction, and request a supplier credit.

2. Problem To Address

- Evidence is incomplete, contradictory, and split across ERP/WMS records, contracts, emails, and operational notes.
- The agent must choose what to inspect next; each observation changes the best next step.
- Several investigation paths can be valid, but querying everything wastes time and cost.
- Enterprise-specific authority, policy, and evidence conventions are absent from generic models.
- The recommendation must be supported, auditable, and constrained by financial controls.

Scope boundary: ERP rules retain matching and controls; solvers optimize only known structured decisions. If facts can be extracted upfront and resolved by rules or a solver, reject this RFT use case.

3. Why RFT + Frontier Tuning

LLM: reconciles contracts, correspondence, warehouse language, and structured transactions.

RFT: learns an investigation policy from outcome reward, not one golden trajectory. Any path can earn reward when it reaches the correct root cause and safe disposition with sufficient evidence and few unnecessary queries.

Frontier Tuning: combines company data, workflows, virtualized tools, eval signals, tuned models, skills, orchestration, and a compliance-aware runtime harness inside the enterprise boundary.

4. Solution Constituents

- **Case router:** routine exceptions to deterministic workflows; ambiguous cases to the agent.
- **Enterprise evidence:** transactions, contracts, quality records, emails, and notes with provenance.
- **Virtualized tools:** read-only ERP, WMS, contract, and correspondence queries in the RLE.
- **Outcome grader:** root cause, safe disposition, evidence sufficiency, efficiency, timeliness, and zero control bypass.
- **Control harness:** RBAC, accounting invariants, separation of duties, abstention, traces, and human approval for writes.

Hill-Climbing Approach

